Limits & Derivatives

Evaluating limits by direct substitution

1 Evaluate $\lim_{x \to 2} (3x^2 - 4x + 1)$.

2 Evaluate $\lim_{x \to -1} (x^3 + 2x - 5)$.

3 Evaluate $\lim_{x \to 0} \frac{5x + 7}{x + 3}$.

4 Evaluate $\lim_{x \to 3} \sqrt{x + 13}$.

Evaluating limits by factoring (the 0/0 form)

5 Evaluate $\lim_{x \to 3} \frac{x^2 - 9}{x - 3}$.

6 Evaluate $\lim_{x \to -2} \frac{x^2 + 5x + 6}{x + 2}$.

7 Evaluate $\lim_{x \to 4} \frac{x^2 - 16}{x - 4}$.

8 Evaluate $\lim_{x \to 1} \frac{x^2 - 1}{x^2 - 3x + 2}$.

Limits at infinity

9 Evaluate $\lim_{x \to \infty} \frac{3x^2 + 2x}{x^2 - 5}$.

10 Evaluate $\lim_{x \to \infty} \frac{2x + 7}{x^2 + 1}$.

11 Evaluate $\lim_{x \to \infty} \frac{5x^3 - 2}{x^3 + 4x}$.

One-sided limits

12 A function is defined by $f(x) = x + 1$ for $x < 2$, and $f(x) = 2x - 1$ for $x \geq 2$. Find $\lim_{x \to 2^-} f(x)$.

13 Using the same piecewise function, find $\lim_{x \to 2^+} f(x)$.

14 Using the same piecewise function, does $\lim_{x \to 2} f(x)$ exist? State its value.

The power rule

15 Differentiate $f(x) = x^5$.

16 Differentiate $f(x) = 3x^4 - 2x^2 + 7$.

17 Differentiate $f(x) = \sqrt{x}$.

18 Differentiate $f(x) = \frac{1}{x^2}$.

The product rule

19 Differentiate $f(x) = x^2(3x + 1)$ using the product rule.

20 Differentiate $f(x) = (x + 2)(x - 5)$ using the product rule.

21 Differentiate $f(x) = (2x - 1)(x^2 + 3)$ using the product rule.

The chain rule

22 Differentiate $f(x) = (3x + 1)^4$.

23 Differentiate $f(x) = (x^2 - 5)^3$.

24 Differentiate $f(x) = \sqrt{4x + 9}$.

25 Differentiate $f(x) = (5 - 2x)^6$.

Derivatives of trigonometric and exponential functions

26 Differentiate $f(x) = \sin x + \cos x$.

27 Differentiate $f(x) = 3\sin x$.

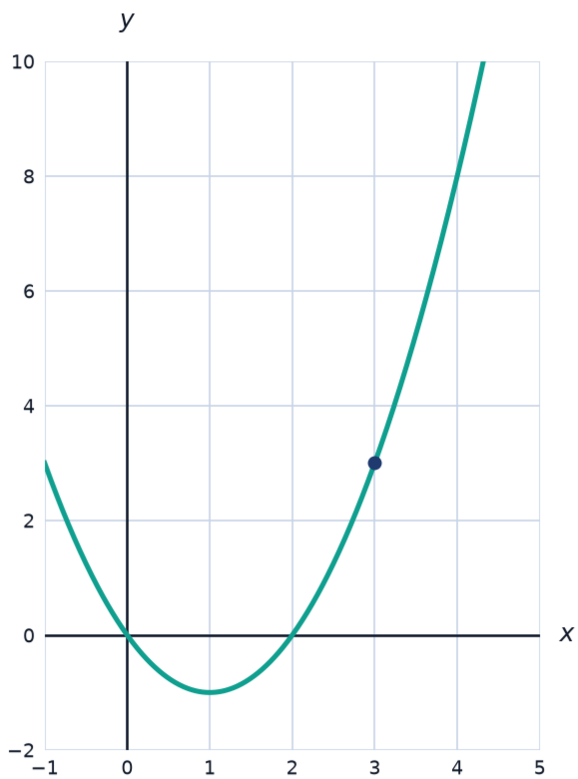
28 Differentiate $f(x) = e^x + x^2$.

Tangent lines

29 Find the slope of the tangent line to $f(x) = x^2$ at $x = 3$.

30 Find the equation of the tangent line to $f(x) = x^2$ at $x = 2$.

31 The graph of $f(x) = x^2 - 2x$ is shown. Find the equation of the tangent line at $x = 3$ (the marked point).



32 Find the x-coordinate(s) where the tangent to $f(x) = x^3 - 3x$ is horizontal.

Increasing, decreasing, and the second derivative

33 For $f(x) = x^2 - 4x + 1$, find $f'(x)$ and determine the interval where f is increasing.

34 For $f(x) = -x^2 + 6x$, determine the interval where f is decreasing.

35 Find the second derivative of $f(x) = x^4 - 2x^2$.

Word problems: rates of change

36 This question has two parts. The height of a ball thrown upward is given by $h(t) = -5t^2 + 20t + 2$ (in meters, t in seconds).

a Find the velocity function $h'(t)$.

b Find the velocity at $t = 1$ second, and state whether the ball is rising or falling.

37 This question has two parts. A company's cost to produce x items is $C(x) = 0.01x^2 + 5x + 200$ (in dollars).

a Find the marginal cost function $C'(x)$.

b Find the marginal cost at $x = 100$ items, and interpret its meaning.